

ANESU RIRWA

Data & AI Engineer | Data Systems, Business Intelligence & Applied Machine Learning

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PROFESSIONAL SUMMARY

Data & AI Engineer and Technology Lead at Munisa Resources, a Zimbabwean mining and infrastructure group. Designed and built the group's in-house ERP and an online mineral-asset marketplace, and moved operations from paper logbooks and scattered spreadsheets onto a single system with automated reporting and forecasting. Works across the full stack: data capture, PostgreSQL database design, Power BI dashboards, predictive models and production web applications in Next.js and TypeScript. Also runs Kordel Data, delivering data systems for mining and healthcare clients. BSc (Hons) in Artificial Intelligence and Machine Learning; pursuing Microsoft Azure certification.

EXPERIENCE

Munisa Resources

Harare, Zimbabwe

Group holding company with subsidiaries in mining (Munisa Mines and Minerals), civil infrastructure (Kenmil Civil Infrastructure) and commodities (Terrafield Commodities).

Data & AI Engineer and Technology Lead

November 2025 – Present

- Own the technology stack and innovation roadmap for Munisa Resources and its three subsidiaries, from architecture decisions through build, deployment and support.
- Designed, built and deployed **MRIS (Munisa Resources Internal System)**, a full-stack ERP for group-wide mining and corporate operations, using Next.js, TypeScript and PostgreSQL (Supabase).
- Built the MRIS central relational database and AI-driven reporting modules that parse daily shift inputs, generate production summaries and forecast operational trends.
- Lead data analysis and business intelligence for the group, producing weekly, monthly and board-level reports on operations, production metrics and market trends.
- Consolidated group records into one system with live dashboards used by directors, finance, procurement, the mine manager and external consultants.
- Built and launched **DealRoom** (dealroom.munisaresources.com), a Next.js marketplace connecting Zimbabwean mineral claim holders with buyers and investors, with confidential teaser listings, buyer and investor mandate registration, and title details released only under NDA.
- Rebuilt the group's three corporate websites (munisaresources.com, kenmilcivilinfrastructure.com, terrafieldcommodities.com) and improved their search engine visibility.

Data Analyst

July 2025 – October 2025

- Digitized legacy Word documents and physical logbooks into an interim Excel and Power BI tracking system, keeping reporting running while the permanent ERP was built.
- Centralized company records in SharePoint, replacing isolated spreadsheets and creating the data backend for the first live dashboards.
- Automated tracking of explosives usage, production output and labour metrics, cutting manual data preparation time by 70%.
- Built a Python machine learning model forecasting production output and operational costs ($R^2 = 0.82$) to support operational and budget planning.

Kordel Data · kordeldata.com

Harare, Zimbabwe

Founder and Principal Consultant

January 2026 – Present

Independent data and AI engineering practice serving mining, industrial and healthcare clients.

- **Maphompe Minerals (March 2026 – Present):** designed and implemented a structured data capture system from scratch, standardizing operational data collection and creating a reliable backend for reporting.
- Built and deployed live interactive dashboards giving Maphompe stakeholders day-to-day visibility of operational performance.
- **Capital Physiotherapy (August 2026 – Present):** building the clinic's digital presence from scratch, starting with a public website and an internal system for patient records and bookings.
- Designing patient-record storage to comply with Zimbabwe's Cyber and Data Protection Act, including role-based access, encryption and patient consent.
- Designing the clinic's data architecture as the foundation for future analytics and machine learning once enough patient and booking data is collected.

Gifted Community Organisation

Harare, Zimbabwe

Data Analyst & Software Engineer

January 2025 – June 2026

- Built the organisation's website to improve donor visibility, engagement and transparency.
- Researched funding models, investment use and resource allocation to guide the organisation's growth and financial planning.

Webgems

Harare, Zimbabwe

Software Engineer Intern

August 2022 – August 2023

- Built and delivered responsive websites and cross-platform mobile apps using React, WordPress and Ionic.
- Wrote tutorials and onboarding documentation for new team members and application users.

TECHNICAL SKILLS

Programming: Python, TypeScript, SQL

Machine learning: scikit-learn, XGBoost, Random Forest, regression and forecasting, NLP, TensorFlow, PyTorch

Data & BI: PostgreSQL, Supabase, relational database design, ETL pipelines, Pandas, NumPy, Power BI, Tableau, Excel, SharePoint

Software: Next.js, React, Flask, Git, HTML/CSS, SEO, WordPress, Ionic

EDUCATION

University of Zimbabwe

August 2020 – July 2024

BSc (Honours) Artificial Intelligence and Machine Learning · Upper Second Class (2.1)

- Thesis: Optimising customer service operations for SMEs and solopreneurs using artificial intelligence and machine learning.
- Class Representative.

PROJECTS

Mining Block Model: Ore Grade Prediction *(in progress)*

- Building XGBoost and Random Forest models to predict the ore grade of unmined blocks and flag economically viable blocks, currently trained on a public Kaggle open-pit block model (75,000 blocks with coordinates, rock type, grade and extraction cost).
- Compiling a comprehensive dataset of Zimbabwean mining data to retrain and validate the models on local deposits.
- Engineering spatial features such as neighbouring-block grade averages, with a dashboard showing how site managers would use the predictions in mine planning.

CERTIFICATIONS

- Python for Data Science — IBM (2025)
- Microsoft Azure Fundamentals (AZ-900) — in progress, exam November 2026
- Microsoft Certified: Azure AI Engineer Associate (AI-102) — preparing, from November 2026

COMMUNITY

- **Uncommon.org** — Volunteer Mentor (November 2023 – May 2025): mentored students in programming and data science and advised them on securing tech internships.
- **All In Open Source** — All In Africa Program (November 2023 – February 2024): open-source collaboration through hackathons and technical workshops.

LANGUAGES

English — Duolingo English Test: 140 **Shona** — Fluent **German** — A1 (score 78); A2 in progress